



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to VELLAVAN SAKKTHI FILLING

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

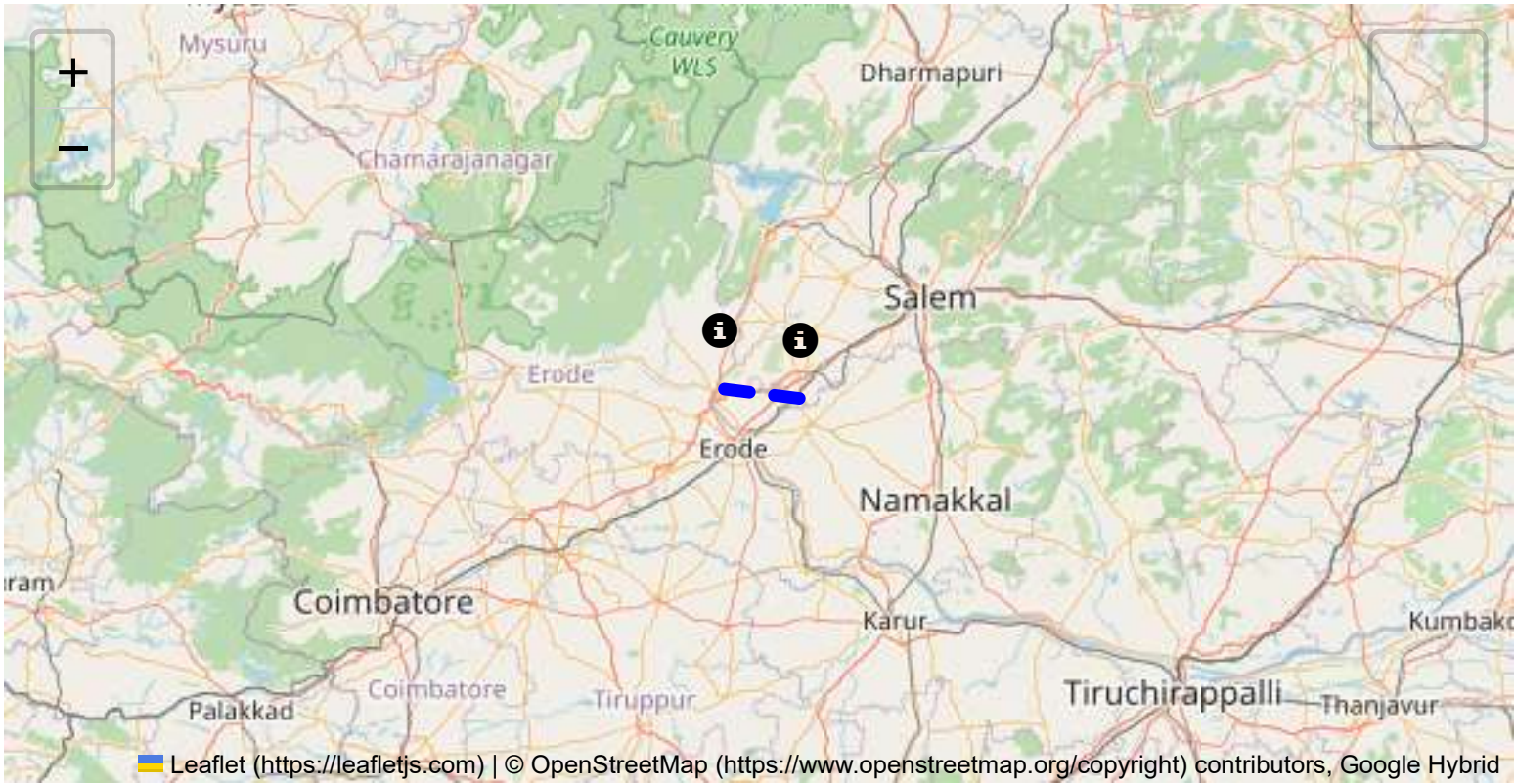
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

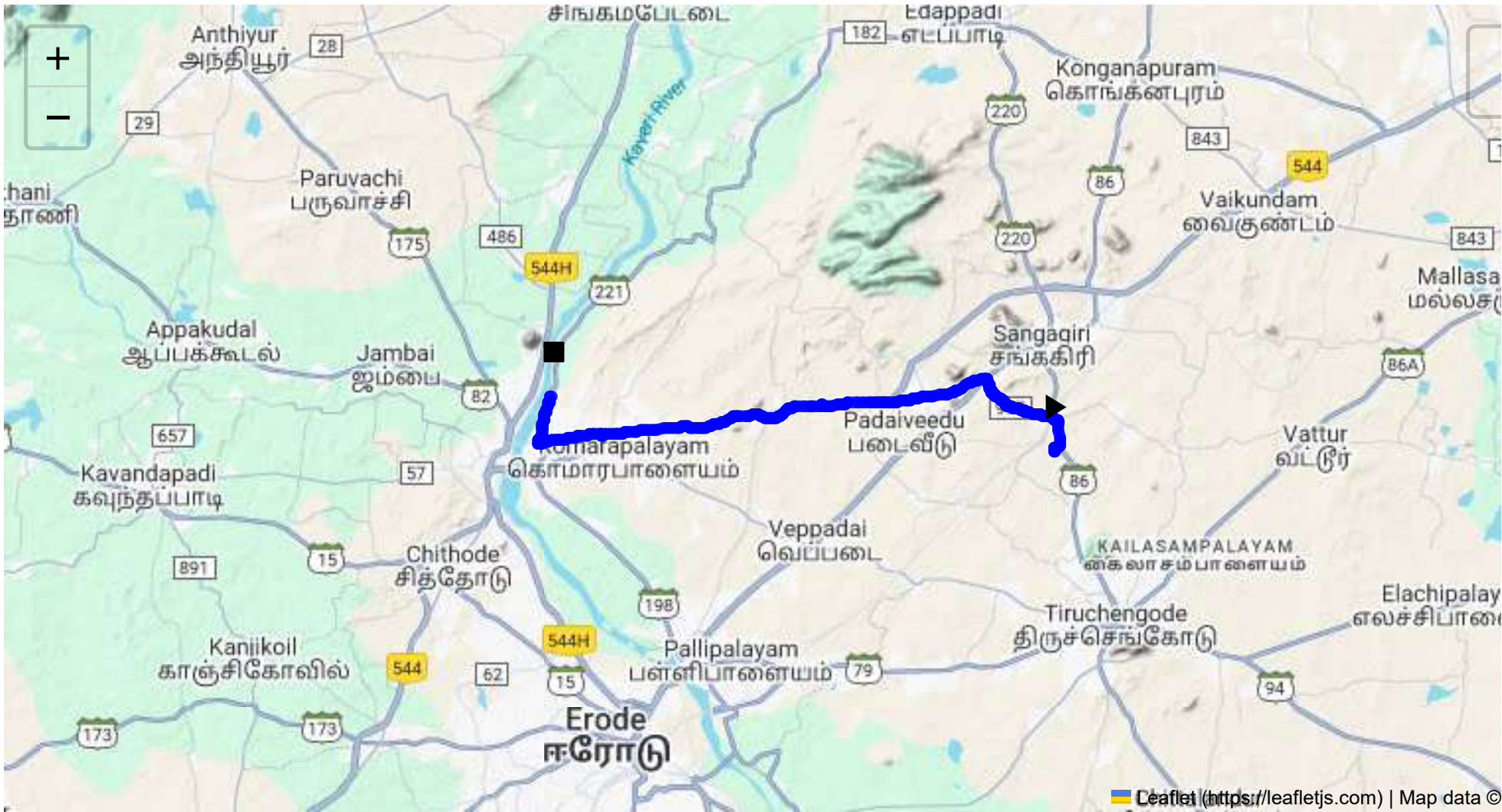
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 24.22 km
Estimated Duration: 0.6 hours
Adjusted Duration (Heavy Vehicle): 0.7 hours
Start: (11.4381, 77.8734)
End: (11.457034, 77.69788)



Welcome to the Journey Risk Management Study

Route Safety Analysis for Sangagiri to Komarapalayam, Tamil Nadu

1. Overview of the Route Map: The route covers approximately 24.22 kilometers, beginning at Sangagiri and ending at the Sri Sakthivinayagha Nursery and Primary School in Komarapalayam. The trip predominantly follows regional roads that might pass through small towns and agricultural areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu generally experiences tropical weather characterized by a hot and humid climate, with monsoon season from June to September. Potential hazards include:

- **Heavy rainfall** leading to waterlogging or reduced visibility.
- **Fog** in colder months, particularly in early mornings.
- **High temperatures** causing road surface deterioration or driver fatigue.

3. Traffic Patterns:

- **Peak Hours:** Morning (8 AM - 10 AM) and evening rush (5 PM - 7 PM) can witness increased congestion, especially near urban settlements.
- **Congestion-Prone Areas:** Market areas and junctions in Sangagiri and Komarapalayam might have notable traffic slowdowns.

4. Assessment of Road Quality and Infrastructure:

- Varying road quality with potential potholes, especially post-monsoon.
- Narrow sections in villages or areas under maintenance can cause bottlenecks.
- Inadequate signage or lighting in less urbanized sections.

5. Suggestions for Alternative Routes for Emergencies:

- Using National Highway (NH) routes as alternatives for better road quality and infrastructure.
- Local bypasses around congested areas, where available, should be identified and marked.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Strict adherence to hazardous materials transport permits.
- Restricted times for transport through densely populated areas.
- Compliance with local and national safety standards for vehicle conditions.

7. Overview of Historical Incidents: There's limited publicly available data on historical incidents specific to this route. However, the general risk is from traffic accidents involving heavy vehicles often attributed to road conditions, driver error, and inadequate maintenance.

8. Environmental Considerations and Sensitive Areas:

- Areas close to water bodies or agricultural lands requiring cautious navigation to prevent contamination.
- Near schools or hospitals, adherence to reduced noise levels and emissions is recommended.

9. Analysis of Communication Coverage:

- Reasonable communication coverage along the route, with potential dead zones in rural stretches.
- Noted more significant risk for coverage loss near agricultural expanses and away from major towns.

10. Estimated Emergency Response Times:

- Urban areas like Sangagiri and Komarapalayam: 15-30 minutes.

- Rural stretches: 45 minutes to an hour, depending on proximity to the nearest emergency services.

11. Overall Summary of Risk Assessment: The route involves moderate risk primarily due to inadequate road conditions, variable traffic congestion, and weather-related disruptions. Minor routes through towns necessitate careful navigation to avoid congestion and comply with local transport regulations for hazardous materials. Advanced planning for emergencies and communication challenges is crucial, with route selections favoring infrastructure supporting quicker responses.

Conclusion: The route from Sangagiri to Komarapalayam should be manageable with judicious preparation focusing on weather forecasts, traffic updates, and route knowledge. Regular updates on road conditions and transport regulations are essential for safe travel.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.46307, 77.84941	30 KM/Hr
5	Turn	Medium	11.46318, 77.84913	30 KM/Hr
6	Turn	High	11.45517, 77.81345	15 KM/Hr
7	Turn	High	11.45521, 77.81343	15 KM/Hr
8	Turn	Medium	11.44467, 77.72092	30 KM/Hr
9	Turn	High	11.44377, 77.71562	15 KM/Hr
10	Blind Spot	Blind Spot	11.44105, 77.69331	10 KM/Hr
11	Turn	High	11.45711, 77.69765	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital	11.4525596, 77.7749426	30 km/h	Medium
4	hospital	Dhanvantri Multi Speciality Hospital	11.451362, 77.766602	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
5	hospital	Dhanvanthri Hospital	11.4496712, 77.7593772	30 km/h	Medium
6	hospital	J.K.K. Trust Hospital	11.4445841, 77.7307962	30 km/h	Medium
11	hospital	Senthil Nursing Home	11.4395054, 77.6957293	30 km/h	Medium
12	hospital	Govt hospital	11.4432595, 77.6941951	30 km/h	Medium
13	hospital	Government Hospital, Komarapalayam	11.4435826, 77.6946365	30 km/h	Medium
14	police	Komarapalayam Police station	11.4413141, 77.6931115	30 km/h	Medium
15	pharmacy	Devi pharmacy	11.4410349, 77.6928556	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
1	college	Vivekanandha Engineering College	11.4589312, 77.7899284	30 km/h	Medium
2	marketplace	Monday market	11.452863, 77.775989	30 km/h	Medium
7	school	JKK Matriculation and Higher Secondary school	11.4446843, 77.7066207	30 km/h	Medium
8	school	JKK Girls Higher Secondary school	11.4437747, 77.7066298	30 km/h	Medium
9	college	Government College of Education	11.4476704, 77.705553	30 km/h	Medium
10	school	Government Girls Hr Sec School	11.4456929, 77.7007497	30 km/h	Medium
16	school	Municipality Middle School	11.4386238, 77.6957329	30 km/h	Medium

Route Photos of Risky Spots

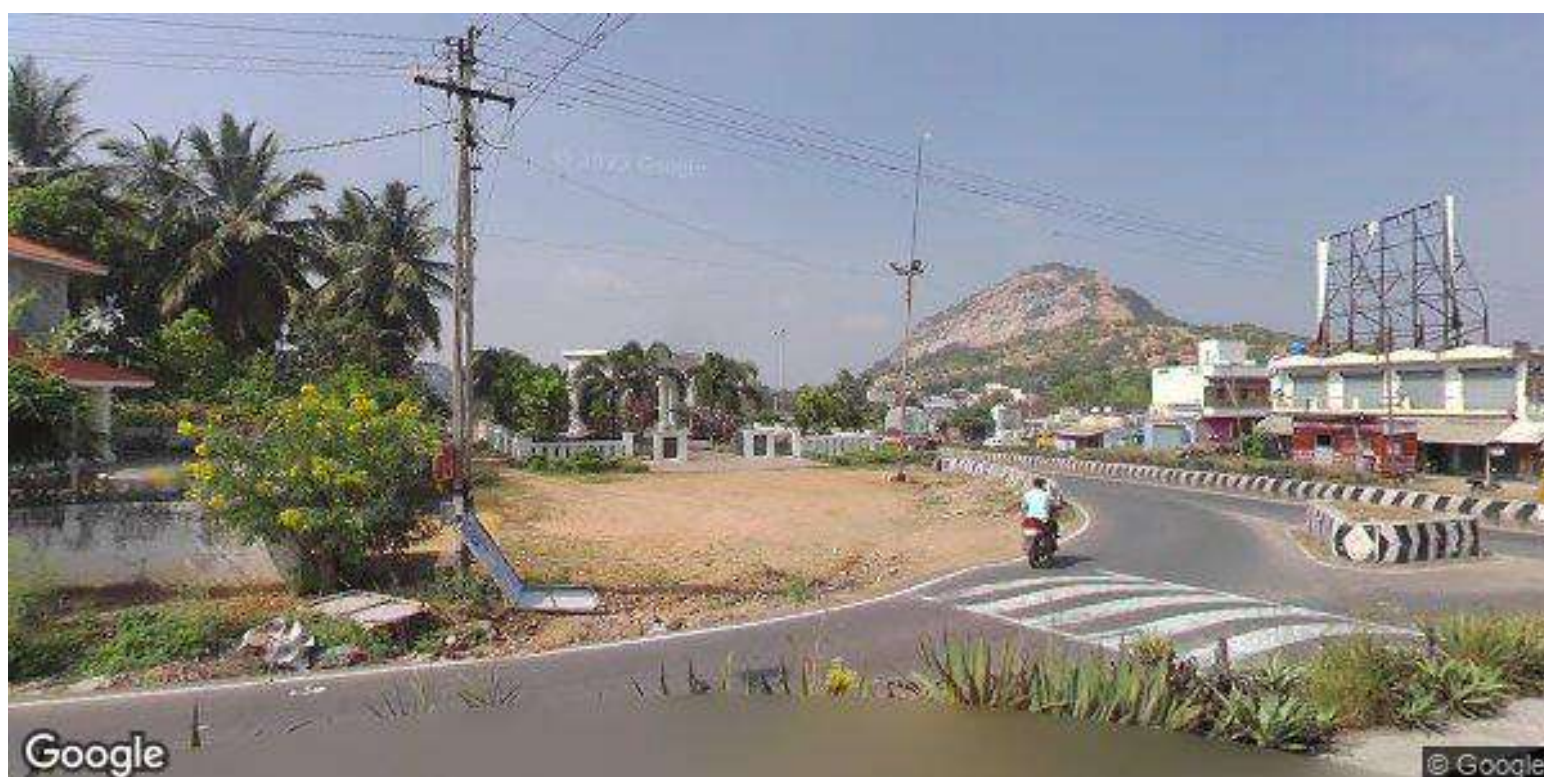


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46307, 77.84941



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45517, 77.81345



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45521, 77.81343



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44467, 77.72092



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44377, 77.71562



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44105, 77.69331



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45711, 77.69765

